Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 6: If and Only If

The Biconditional

Chartrand & Zhang, Section 1.4

Name: _____ Date: _____

Where we are

Wednesday you found that an implication and its converse are different statements. Sometimes, though, *both* hold. That situation is common enough — and important enough in mathematics — to get its own connective, which is today's topic. It is also the last connective in Chapter 1.

By the end of today you should be able to

1. State the truth table for $P \Leftrightarrow Q$.
2. Recognise "if and only if" and "necessary and sufficient" as the same thing.
3. Decide whether a biconditional is true for a given pair of statements.
4. Explain why proving a biconditional means proving two implications.

1. The definition

Definition 1.52 — biconditional

For statements P and Q, the **biconditional** of P and Q is

$$P \Leftrightarrow Q \;:=\; (P \Rightarrow Q) \wedge (Q \Rightarrow P)$$

read "P **if and only if** Q," or "P **is necessary and sufficient for** Q."

Since it is defined as a conjunction of things you already know, you can build its table yourself.

P	Q	$P \Rightarrow Q$	$Q \Rightarrow P$	$P \Leftrightarrow Q$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

$P \Leftrightarrow Q$ is true exactly when P and Q have the same truth value.

Where the name comes from

Recall from Wednesday that $Q \Rightarrow P$ can be read "P *if* Q," and $P \Rightarrow Q$ can be read "P *only if* Q." Their conjunction is therefore

$$\text{“}P \text{ if } Q\text{”} \ \wedge \ \text{“}P \text{ only if } Q\text{”} \quad = \quad \text{“}P \text{ if and only if } Q\text{.”}$$

The phrase is not an idiom. It is literally the two halves stated together. Mathematicians abbreviate it **iff**. Not a word, but you will see it constantly.

2. An old friend in disguise

Compare the $P \Leftrightarrow Q$ column with the $P \oplus Q$ column from Week 1.

P	Q	$P \Leftrightarrow Q$	$P \oplus Q$	$\sim(P \oplus Q)$
T	T	T	F	T
T	F	F	T	F
F	T	F	T	F
F	F	T	F	T

$$P \Leftrightarrow Q \ \equiv \ \sim(P \oplus Q) \qquad \text{and equivalently} \qquad \sim(P \Leftrightarrow Q) \ \equiv \ P \oplus Q$$

You built this in August

Homework 2, Problem 1(b) asked for the table of $\sim(P \oplus Q)$. That was the biconditional, three weeks before it had a name. Exclusive or asks whether the two *differ*; the biconditional asks whether they *agree*. They are negations of each other.

3. Deciding whether a biconditional is true

Only one question matters: **do the two sides have the same truth value?**

Biconditional	P	Q	$P \Leftrightarrow Q$
7 is prime if and only if 9 is composite.	T	T	True
$3^2 > 2^3$ if and only if $4^3 > 3^4$.	T	F	False
5 is even if and only if 6 is odd.	F	F	True

The third row is the one that unsettles people

"5 is even if and only if 6 is odd" is **true**, and both halves are false. That is correct: two false statements agree with each other, so the biconditional holds.
As with the implication, the definition looks only at truth values. It does not require the two sides to be about the same subject, or to be related at all.

4. Biconditionals with open sentences

For an integer n, let $P(n)$: n is odd, and $Q(n)$: n^2 is odd.

n	$P(n)$	$Q(n)$	$P(n) \Leftrightarrow Q(n)$
3	T	T	T
4	F	F	T
-7	T	T	T

It appears that $P(n) \Leftrightarrow Q(n)$ is true for every integer n.

Why this is the shape of a real theorem

"An integer is odd if and only if its square is odd" is a genuine theorem, and we will prove it in Chapter 3.
To prove it you must prove **two** things: if n is odd then n^2 is odd, *and* if n^2 is odd then n is odd. A biconditional is two implications, so a proof of one is half a proof. Forgetting the second half is the most common way an "iff" proof goes wrong.

The six connectives — Chapter 1 complete

Symbol	Name	True when
$\sim P$	negation	P is false
$P \wedge Q$	conjunction	both are true
$P \vee Q$	disjunction	at least one is true
$P \oplus Q$	exclusive or	exactly one is true
$P \Rightarrow Q$	implication	*not* (P true and Q false)
$P \Leftrightarrow Q$	biconditional	both have the same truth value

Only two are really needed

Homework 2's challenge showed $P \wedge Q \equiv \sim((\sim P) \vee (\sim Q))$. Theorem 1.48 gives $P \Rightarrow Q \equiv (\sim P) \vee Q$. Definition 1.52 builds $\Leftrightarrow$ from $\Rightarrow$ and $\wedge$.
Chase it through and every connective in the table can be written using only $\sim$ and $\vee$. The other four exist for readability, not necessity — which is also why hardware can be built entirely out of

one kind of gate.

Exit ticket

1. Is "$2 + 2 = 5$ if and only if $3 > 7$" true or false? Explain in one line.

2. What two things must you prove to establish "n is even if and only if n^2 is even"?

Before Monday

Read Section 1.5, "Tautologies and Contradictions," beginning p. 40.

Self-check (not collected). Section 1.4, Exercises 1, 3, and 5 — odd-numbered, answers in the back. Attempt them fully first.

Coming up

Homework 2 is due tonight, Friday, August 28, 11:59 p.m.
Quiz 2 is Monday, August 31, in class — 15 minutes on Section 1.3 (implications, converse, contrapositive).
Homework 3 covers Sections 1.3 and 1.4 and is due Friday, September 4.

Questions I still have